Flexible $p(\chi_{\rm eff}, \chi_p \mid m_1, q)$

all q | $q \in [0.8, 1.0]$ | $q \in [0.3, 0.65]$

marginal, median and 90%

$P(q \in Q)$ $0.6^{+0.19}_{-0.42}$

$P(q \in Q)$ $0.16^{+0.16}_{-0.077}$

all m_1

$m_1 \in [8, 12]\, M_\odot$

$P(m_1 \in M)$ $0.66^{+0.093}_{-0.13}$

$P(q \in Q \mid m_1 \in M)$ $0.72^{+0.21}_{-0.6}$

$P(q \in Q \mid m_1 \in M)$ $0.055^{+0.21}_{-0.05}$

$m_1 \in [16, 22]\, M_\odot$

$P(m_1 \in M)$ $0.069^{+0.05}_{-0.032}$

$P(q \in Q \mid m_1 \in M)$ $0.076^{+0.21}_{-0.067}$

$P(q \in Q \mid m_1 \in M)$ $0.65^{+0.22}_{-0.26}$

$m_1 \in [30, 40]\, M_\odot$

$P(m_1 \in M)$ $0.046^{+0.022}_{-0.016}$

$P(q \in Q \mid m_1 \in M)$ $0.74^{+0.16}_{-0.22}$

$P(q \in Q \mid m_1 \in M)$ $0.057^{+0.098}_{-0.046}$

$m_1 \in [55, 65]\, M_\odot$

$P(m_1 \in M)$ $0.0045^{+0.0061}_{-0.0026}$

$P(q \in Q \mid m_1 \in M)$ $0.039^{+0.18}_{-0.036}$

$P(q \in Q \mid m_1 \in M)$ $0.67^{+0.23}_{-0.38}$

χ_p

$\chi_{\rm eff}$

$p(\chi_{\rm eff}, \chi_p \mid m_1, q)$